EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Units and Dimensions

1. In a book, the answer for a particular question is expressed as $b = \frac{ma}{k}\left[\sqrt{1+\frac{2k\ell}{ma}}\right]$ here m represents mass, a represents acceleration, ℓ represents length. The unit of b should be :-

(A) m/s (B) m/s^2 (C) meter (D) $/sec$

PUV084

2. A particle is in a unidirectional potential field where the potential energy (U) of a particle depends on the x-coordinate given by $U_x = k(1 - \cos ax)$ and k and 'a' are constants. Find the dimension of 'a' and 'k'.

(A) $[L]$, $[MLT^{-2}]$ (B) $[L^{-1}]$, $[ML^2T^{-2}]$ (C) $[L^{-1}]$, $[ML^{-1}T^{-2}]$ (D) None of these

PUV086

3. The equation of the stationary wave is $y = 2A\sin\left(\frac{2\pi ct}{\lambda}\right)\cos\left(\frac{2\pi x}{\lambda}\right)$ which of the following statement is wrong?

(A) The unit of ct is same as that of λ (B) The unit of x is same as that of λ

(C) The unit of $\frac{2\pi c}{\lambda}$ is same as that of $\frac{2\pi x}{\lambda t}$ (D) The unit of $\frac{c}{\lambda}$ is same as that of $\frac{x}{\lambda}$

PUV093

4. Which of the following can be a set of fundamental quantities

(A) length, velocity, time (B) momentum, mass, velocity

(C) force, mass, velocity (D) momentum, time, frequency

PUV081

5. If area (A), velocity (v), and density (ρ) are base units, then the dimensional formula of force can be represented as :-

(A) $Av\rho$ (B) $Av^2\rho$ (C) $Av\rho^2$ (D) $A^2v\rho$

PUV090

6. In a certain system of units, 1 unit of time is 5 sec, 1 unit of mass is 20 kg and 1 unit of length is 10 m. In this system, one unit of power will correspond to :-

(A) 16 watts (B) $\frac{1}{16}$ watts (C) 25 watts (D) None of these

PUV094

7. Density of wood is 0.5 g/cc in the CGS system of units. The corresponding value in MKS units is :-

(A) 500 (B) 5 (C) 0.5 (D) 5000

PUV091

Basic Mathematics

8. The value of $\frac{\pi\cos 60^\circ}{9\sin 2^0}$ to nearest integer is approximately.

(A) 5 (B) 10 (C) 15 (D) 20

PUV099

For More Join: @IITJEE_Advanced

9. Calculate the approximate value of $(8.24)^{1/3}$.

(A) 1.01 (B) 2.02 (C) 3.03 (D) 4.04

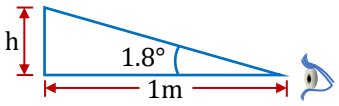
PUV100

10. Use the approximation $(1 + x)^n \approx 1 + nx$, $|x| << 1$, value of $\sqrt{97}$ will be :-

(A) 9.75 (B) 9.80 (C) 9.85 (D) 9.90

PUV101

11. A normal human eye can see an object making an angle of 1.8° at the eye. What is the approximate height of object which can be seen by an eye placed at a distance of 1 *m* from the eye.



(A) $\pi\, cm$ (B) $\frac{\pi}{2}\, cm$ (C) $\frac{\pi}{3}\, cm$ (D) $\frac{\pi}{4}\, cm$

PUV105

12. The approximate value of x where $x = \sin 2° \cos 2°$, is

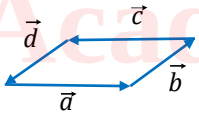
(A) $\frac{\pi}{90}$ (B) 2 (C) 1 (D) $\frac{\pi}{45}$

PUV107

Vectors

13. On the basis of given diagram, choose the pair of vectors which have an acute angle between them:

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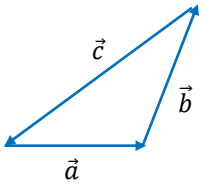


(A) $\vec{a}$ and $\vec{c}$ (B) $\vec{a}$ and $\vec{d}$ (C) $\vec{b}$ and $\vec{c}$ (D) $\vec{c}$ and $\vec{d}$

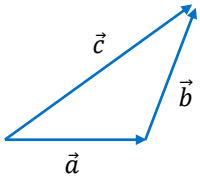
PUV113

14. The vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are related by $\vec{c} = \vec{a} + \vec{b}$. Which diagram below illustrates this relationship?

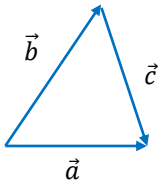
(A)



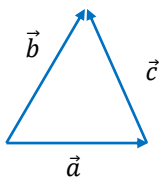
(B)



(C)



(D)



PUV116

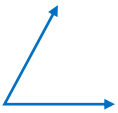
15. If $\vec{A} + \vec{B} = \vec{C}$ and $A = B + C$, then the angle between $\vec{A}$ and $\vec{B}$ is :-

(A) 0 (B) $\frac{\pi}{4}$ (C) π (D) $\frac{\pi}{2}$

PUV117

For More Join: @IITJEE_Advanced

16. Refer the following arrangements consisting of two vectors of same magnitude. Arrange them in ascending order of resultant magnitudes.

(I)  (II)  (III)  (IV) 

(A) I, II, III, and IV (B) IV, III, II and I (C) II, IV, III and I (D) II, I, III and IV

PUV118

17. A man moves towards 3 m north then 4 m towards east and finally 5 m towards 37° south of west. His displacement from origin is :-

(A) $5\sqrt{2}m$ (B) $0m$ (C) 1 m (D) 12 m

PUV119

18. If $\vec{a}+\vec{b}=\vec{c}$, $|\vec{a}|=20$, $|\vec{b}|=20$ and $|\vec{c}|=20\sqrt{3}$ then angle between $\vec{a}$ and $\vec{b}$ is :-

(A) 60° (B) 30° (C) 45° (D) 120°

PUV122

19. A particle is moving westward with a velocity $\vec{v}_1 = 5\ m/s$. Its velocity changed to $\vec{v}_2 = 5\ m/s$ northward. The change in velocity vector $\left(\Delta\vec{V}=\vec{v}_2-\vec{v}_1\right)$ is:

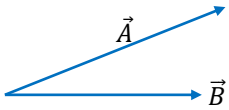
(A) $5\sqrt{2}\ m/s$ towards north east (B) 5 m/s towards north west

(C) zero (D) $5\sqrt{2}\ m/s$ towards north west

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PUV124

20. Two vectors $\vec{A}$ and $\vec{B}$ have magnitudes 2 and 1 respectively. If the angle between $\vec{A}$ and $\vec{B}$ is 60°, which of the following vectors may be equal to $\frac{\vec{A}}{2}-\vec{B}$.



(A)  (B)  (C) (D)

PUV125

21. The resultant of two forces, one double the other in magnitude is perpendicular to the smaller of the two forces. The angle between two forces is :-

(A) 150° (B) 90° (C) 60° (D) 120°

PUV126

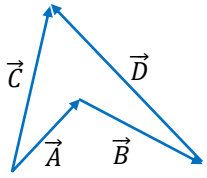
22. Two vectors $\vec{a}$ and $\vec{b}$ add to give a resultant $\vec{c}=\vec{a}+\vec{b}$. In which of these cases angle between $\vec{a}$ and $\vec{b}$ is maximum: (a, b, c represent the magnitudes of respective vectors)

(A) $c = a + b$ (B) $c^2 = a^2 + b^2$ (C) $c = a - b$ (D) can not be determined

PUV129

For More Join: @IITJEE_Advanced

23. Which of the following is correct regarding these four vectors?



(A) $\vec{A}+\vec{B}+\vec{C}+\vec{D}=0$ (B) $\vec{C}-\vec{A}-\vec{B}-\vec{D}=0$
(C) $\vec{C}-\vec{A}+\vec{B}-\vec{D}=0$ (D) $\vec{A}+\vec{B}-\vec{C}-\vec{D}=0$

PUV133

24. The ratio of maximum and minimum magnitudes of the resultant of two vectors $\vec{A}$ and $\vec{B}$ is 3 : 2. The relation between A and B is

(A) $A = 5B$ (B) $5A = B$ (C) $A = 3B$ (D) $A = 4B$

PUV134

25. A body placed in free space, is simultaneously acted upon by three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$. The body is in equilibrium and the forces $\vec{F}_1$ and $\vec{F}_2$ are known to be 36 N due north and 27 N due east respectively. Which of the following best describes the force $\vec{F}_3$?

(A) 36 N due south (B) 53 N due 60° south of east
(C) 45 N due 53° south of west (D) 45 N due 37° north of west

PUV137

26. $\vec{A}$ and $\vec{B}$ have magnitudes 5 units and 12 units respectively. If the angle between their vector addition and $\vec{A}$ is numerically equal to the angle between their vector difference and $\vec{A}$, then the resultant of these two vectors ($\vec{A}$ and $\vec{B}$) has the magnitude:

(A) $\sqrt{109}$ units (B) $\sqrt{229}$ units (C) 13 units (D) 7 units

PUV138

27. There are two vectors $\vec{A}=3\hat{\imath}+\hat{\jmath}-2\hat{k}$ and $\vec{B}=4\hat{\imath}-2\hat{\jmath}-6\hat{k}$ Find the unit vector along $\vec{C}=\vec{A}+\vec{B}$

(A) $\frac{7\hat{\imath}-\hat{\jmath}+8\hat{k}}{\sqrt{114}}$ (B) $\frac{7\hat{\imath}-\hat{\jmath}-8\hat{k}}{\sqrt{114}}$ (C) $\frac{7\hat{\imath}+\hat{\jmath}-8\hat{k}}{\sqrt{104}}$ (D) $\frac{7\hat{\imath}-\hat{\jmath}-8\hat{k}}{\sqrt{104}}$

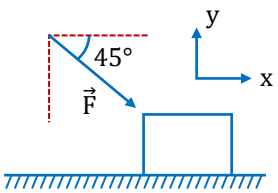
PUV141

28. Position of a particle at t = 0 is (2, 3, 0). It starts moving with a speed of 10 m/s in direction 37° north of west. Its position after t = 1 sec is
(Take North as positive y-axis and East as positive x-axis):

(A) (6, 11) (B) (–4, 11) (C) (–8, 6) (D) (–6, 9)

PUV142

29. A person pushes a box kept on a horizontal surface with force of 100 N. In unit vector notation force $\vec{F}$ can be expressed as :



(A) $100\,(\hat{\imath}+\hat{\jmath})$ (B) $100\,(\hat{\imath}-\hat{\jmath})$ (C) $50\sqrt{2}(\hat{\imath}-\hat{\jmath})$ (D) $50\sqrt{2}(\hat{\imath}+\hat{\jmath})$

PUV146

For More Join: @IITJEE_Advanced

30. The projection of a vector, $\vec{r} = 3\hat{\imath} + \hat{\jmath} + 2\hat{k}$, on the $x-y$ plane has magnitude :–

(A) 3 (B) 4 (C) $\sqrt{13}$ (D) $\sqrt{10}$

PUV147

31. For the given vector $\vec{A} = 3\hat{\imath} - 4\hat{\jmath} + 10\hat{k}$, the ratio of magnitude of its component on the x-y plane and the component on z–axis is :-

(A) 2 (B) $\frac{1}{2}$ (C) 1 (D) None

PUV148

32. The dot product of two vectors of magnitudes 3 units and 5 units cannot be:

(A) 2 (B) –2 (C) 20 (D) Zero

PUV151

33. When $\vec{A} \cdot \vec{B} = -|\vec{A}||\vec{B}|$, then:

(A) $\vec{A}$ and $\vec{B}$ are perpendicular to each other (B) $\vec{A}$ and $\vec{B}$ act in the same direction

(C) $\vec{A}$ and $\vec{B}$ act in the opposite direction (D) $\vec{A}$ and $\vec{B}$ can act in any direction

PUV152

34. When $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$, then $\vec{A}.\vec{B}$ is equal to :-

(A) 0 (B) AB (C) 1 (D) Zero vector

PUV154

35. If $\vec{A} + \vec{B}$ is perpendicular to $\vec{A} - \vec{B}$, then which of the following option must be correct:

(A) $\vec{A}.\vec{B} = 0$ (B) $\vec{A}.\vec{B} \neq 0$ (C) $|\vec{A}| = |\vec{B}|$ (D) $|\vec{A}| \neq |\vec{B}|$

PUV155

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For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

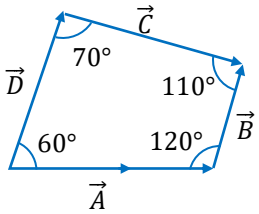
1. For the equation $x = AC \sin(Bt) + De^{(BCt)}$, where x and t represent position and time respectively, which of the following is/are **CORRECT** :-

(A) Dimension of AC is LT^{-1} (B) Dimension of B is T^{-1}

(C) Dimension of AC and D are same (D) Dimension of C is T^{-1}

PUV085

2. Four vector $\vec{A}, \vec{B}, \vec{C}, \vec{D}$ are shown as figure. Then :



(A) Angle between $\vec{A}$ and $\vec{B}$ is 60°. (B) Angle between $\vec{D}$ and $\vec{C}$ is 70°.

(C) Angle between $\vec{A}$ and $\vec{C}$ is 50°. (D) Angle between $\vec{B}$ and $\vec{C}$ is 110°.

PUV114

3. Four forces acting on a particle keep it in equilibrium, then :-

(A) the force must be coplanar.

(B) the forces cannot be coplanar.

(C) the forces may or may not be coplanar.

(D) if three of these forces are coplanar, so must be the fourth.

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PUV131

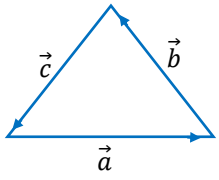
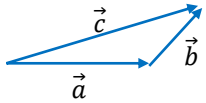
4. Which of the following statements about the sum of the two vectors $\vec{A}$ and $\vec{B}$, is/are correct?

(A) $|\vec{A} + \vec{B}| \leq A + B$ (B) $|\vec{A} + \vec{B}| \geq A + B$ (C) $|\vec{A} + \vec{B}| \geq |\vec{A} - \vec{B}|$ (D) $|\vec{A} + \vec{B}| \geq |A - B|$

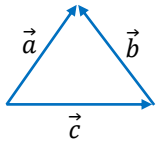
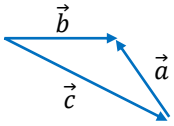
PUV135

MATRIX MATCH TYPE QUESTION

5. Column-I show vector diagram relating three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$. Match the vector equation in column-II, with vector diagram in column-I :

Column-I		**Column-II**	
(A)		(P)	$\vec{a} - (\vec{b} + \vec{c}) = 0$
(B)		(Q)	$\vec{b} - \vec{c} = \vec{a}$

For More Join: @IITJEE_Advanced

Column-I	Column-II
(C) 	(R) $\vec{a}+\vec{b}=-\vec{c}$
(D) 	(S) $\vec{a}+\vec{b}=\vec{c}$

(A) (A)-*R* ; (B)-*S* ; (C)-*P* ; (D)-*Q*
(B) (A)-*P* ; (B)-*Q* ; (C)-*R* ; (D)-*S*
(C) (A)-*R* ; (B)-*P* ; (C)-*Q* ; (D)-*S*
(D) (A)-*P* ; (B)-*S* ; (C)-*R* ; (D)-*Q*

PUV139

6. Column-I contains two vectors whose length represents it's magnitude and arrow represents it's direction. Column-II contains some properties of vector, match them correctly.

Column-I	Column-II
(A) 	(P) Parallel vectors
(B) 	(Q) Antiparallel vectors
(C) 	(R) Equal in magnitude
(D) 	(S) Unequal in magnitude
	(T) Neither parallel nor antiparallel

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(A) (A) P, S (B) Q, R (C) S, T (D) P, R
(B) (A) P, R (B) Q, R (C) S, T (D) P, S
(C) (A) P, R (B) Q, S (C) S, T (D) P, R
(D) (A) P, S (B) P, R (C) S, T (D) P, S

PUV112

For More Join: @IITJEE_Advanced

EXERCISE (S)

Units and Dimensions

1. The equation of state for a real gas at high temperature is given by $P = \frac{nRT}{V-b} - \frac{a}{T^{1/2}V(V+b)}$ where n, P, V and T are number of moles, pressure, volume and temperature respectively and R is the universal gas constant. Find the dimensions of constant a in the above equation.

PUV087

2. Find the value of 'x' in the expression, $A = se^{\frac{at^x}{A}}$ where s : displacement, t : time, a : acceleration

PUV089

3. Energy of a particle is $5J$. If unit of length and time are doubled and unit of mass is halved then numerical value of the energy in this new system will be $5n$. Value of n is.

PUV098

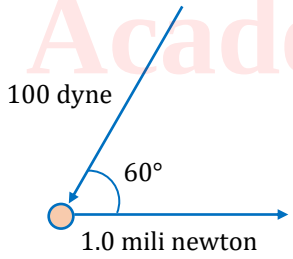
Vectors

4. Two forces of magnitude $5N$ each are applied on a block. One force is acting towards East and the other acting along 60° North of East. The resultant of the two forces (in N) is of magnitude :-

PUV120

5. Two forces act on a particle simultaneously as shown in the figure. Find net force in milli newton on the particle. [Dyne is the *CGS* unit of force]

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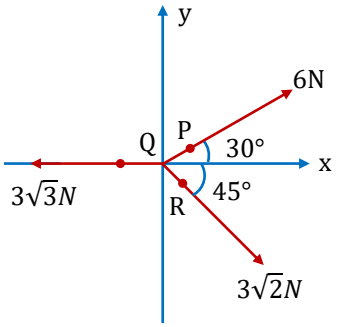


PUV121

6. The maximum and minimum magnitudes of the resultant of two forces are $35N$ and $5N$ respectively. Find the magnitude of resultant force when act orthogonally to each other.

PUV127

7. Three ants P, Q and R are pulling a grain with forces of magnitude $6N$, $3\sqrt{3}N$ and $3\sqrt{2}N$ as shown in the figure. Find the magnitude of resultant force (in N) acting on the grain.



PUV144

For More Join: @IITJEE_Advanced

8. Consider the two vectors : $\vec{L} = 1\hat{\imath} + 2\hat{\jmath} + 3\hat{k}$ and $\vec{l} = 4\hat{\imath} + 5\hat{\jmath} + 6\hat{k}$. Find the value of the scalar α such that the vector $\vec{L} - \alpha\vec{l}$ is perpendicular to $\vec{L}$.

PUV158

9. Find components of vector $\vec{a} = \hat{\imath} + \hat{\jmath} + 3\hat{k}$ in directions parallel to and perpendicular to vector $\vec{b} = \hat{\imath} + \hat{\jmath}$.

PUV159

10. If the velocity of a particle is $(2\hat{\imath} + 3\hat{\jmath} - 4\hat{k})$ and its acceleration is $(-\hat{\imath} + 2\hat{\jmath} + \hat{k})$ and angle between them is $\frac{n\pi}{4}$. The value of n is.

PUV160

11. (a) Calculate $\vec{r} = \vec{a} - \vec{b} + \vec{c}$ where $\vec{a} = 5\hat{\imath} + 4\hat{\jmath} - 6\hat{k}$, $\vec{b} = -2\hat{\imath} + 2\hat{\jmath} + 3\hat{k}$ and $\vec{c} = 4\hat{\imath} + 3\hat{\jmath} + 2\hat{k}$.

(b) Calculate the angle between $\vec{r}$ and the z-axis.

(c) Find the angle between $\vec{a}$ and $\vec{b}$.

PUV161

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For More Join: @IITJEE_Advanced

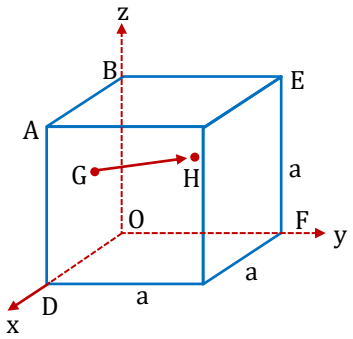
EXERCISE - JEE (Main) PYQ

1. Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. The magnitude of $(\vec{A}+\vec{B})$ is 'n' times the magnitude of $(\vec{A}-\vec{B})$. The angle between $\vec{A}$ and $\vec{B}$ is : **[JEE (Main) 2019]**

(1) $\sin^{-1}\left[\frac{n^2-1}{n^2+1}\right]$ (2) $\cos^{-1}\left[\frac{n-1}{n+1}\right]$ (3) $\cos^{-1}\left[\frac{n^2-1}{n^2+1}\right]$ (4) $\sin^{-1}\left[\frac{n-1}{n+1}\right]$

PUV031

2. In the cube of side 'a' shown in the figure, the vector from the central point of the face *ABOD* to the central point of the face *BEFO* will be: **[JEE (Main) 2019]**



(1) $\frac{1}{2}a(\hat{i}-\hat{k})$ (2) $\frac{1}{2}a(\hat{j}-\hat{i})$ (3) $\frac{1}{2}a(\hat{k}-\hat{i})$ (4) $\frac{1}{2}a(\hat{j}-\hat{k})$

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PUV032

3. Let $|\vec{A}_1|=3, |\vec{A}_2|=5$ and $|\vec{A}_1+\vec{A}_2|=5$. The value of $(2\vec{A}_1+3\vec{A}_2).(3\vec{A}_1-2\vec{A}_2)$ is :- **[JEE (Main) 2019]**

(1) –112.5 (2) –106.5 (3) –118.5 (4) –99.5

PUV033

4. A quantity f is given by $f=\sqrt{\frac{hc^5}{G}}$ where c is the speed of light, G is the universal gravitational constant and h is the Planck's constant. Dimension of f is that of : **[JEE (Main) 2020]**

(1) Momentum (2) Area (3) Energy (4) Volume

PUV034

5. If momentum (P), area (A) and time (T) are taken to be the fundamental quantities then the dimensional formula for energy is : **[JEE (Main) 2020]**

(1) $[PA^{-1}T^{-2}]$ (2) $[PA^{1/2}T^{-1}]$ (3) $[P^2AT^{-2}]$ (4) $[P^{1/2}AT^{-1}]$

PUV035

6. Amount of solar energy received on the Earth's surface per unit area per unit time is defined as solar constant. The dimension of solar constant is: **[JEE (Main) 2020]**

(1) ML^2T^{-2} (2) MLT^{-2} (3) $M^2L^0T^{-1}$ (4) ML^0T^{-3}

PUV036

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7. If time (t), velocity (v), and angular momentum (ℓ) are taken as the fundamental units. Then the dimension of mass (m) in terms of t, v and ℓ is : **[JEE (Main) 2021]**

(1) $[t^{-1} v^{1} \ell^{-2}]$ (2) $[t^{1} v^{2} \ell^{-1}]$ (3) $[t^{-2} v^{-1} \ell^{1}]$ (4) $[t^{-1} v^{-2} \ell^{1}]$

PUV037

8. The force is given in terms of time t and displacement x by the equation

$F = A \cos Bx + C \sin Dt$

The dimensional formula of $\frac{AD}{B}$ is : **[JEE (Main) 2021]**

(1) $[M^{0} L T^{-1}]$ (2) $[M L^{2} T^{-3}]$ (3) $[M^{1} L^{1} T^{-2}]$ (4) $[M^{2} L^{2} T^{-3}]$

PUV038

9. If E, L, M and G denote the quantities as energy, angular momentum, mass and the universal constant of gravitation respectively, then the dimensions of P in the formula $P = EL^{2}M^{-5}G^{-2}$ are: **[JEE (Main) 2021]**

(1) $[M^{0} L^{1} T^{0}]$ (2) $[M^{-1} L^{-1} T^{2}]$ (3) $[M^{1} L^{1} T^{-2}]$ (4) $[M^{0} L^{0} T^{0}]$

PUV039

10. If force (F), length (L) and time (T) are taken as the fundamental quantities, then what will be the dimension of density : **[JEE (Main) 2021]**

(1) $[FL^{-4} T^{2}]$ (2) $[FL^{-3} T^{2}]$ (3) $[FL^{-5} T^{2}]$ (4) $[FL^{-3} T^{3}]$

PUV040

11. Match List-I with List-II.

	List-I		**List-II**
(a)	Torque	(i)	MLT^{-1}
(b)	Impulse	(ii)	MT^{-2}
(c)	Tension	(iii)	$ML^{2}T^{-2}$
(d)	Surface Tension	(iv)	MLT^{-2}

Choose the most appropriate answer from the option given below: **[JEE (Main) 2021]**

(1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii) (2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(3) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii) (4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

PUV041

12. If velocity [V], time [T] and force [F] are chosen as the base quantities, the dimensions of the mass will be : **[JEE (Main) 2021]**

(1) $[FT^{-1} V^{-1}]$ (2) $[FTV^{-1}]$ (3) $[FT^{2} V]$ (4) $[FVT^{-1}]$

PUV042

13. If $\vec{P} \times \vec{Q} = \vec{Q} \times \vec{P}$, the angle between $\vec{P}$ and $\vec{Q}$ is θ ($0° < \theta < 360°$). The value of 'θ' will be _____°. **[JEE (Main) 2021]**

PUV043

14. If $\vec{A}$ and $\vec{B}$ are two vectors satisfying the relation $\vec{A}.\vec{B} = |\vec{A} \times \vec{B}|$. Then the value of $|\vec{A} - \vec{B}|$ will be : **[JEE (Main) 2021]**

(1) $\sqrt{A^2 + B^2}$ (2) $\sqrt{A^2 + B^2 + \sqrt{2}AB}$

(3) $\sqrt{A^2 + B^2 + 2AB}$ (4) $\sqrt{A^2 + B^2 - \sqrt{2}AB}$

PUV045

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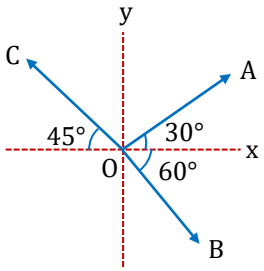
For More Join: @IITJEE_Advanced

15. What will be the projection of vector $\vec{A} = \hat{\imath} + \hat{\jmath} + \hat{k}$ on vector $\vec{B} = \hat{\imath} + \hat{\jmath}$? **[JEE (Main) 2021]**

(1) $\sqrt{2}(\hat{\imath} + \hat{\jmath} + \hat{k})$ (2) $2(\hat{\imath} + \hat{\jmath} + \hat{k})$ (3) $\sqrt{2}(\hat{\imath} + \hat{\jmath})$ (4) $(\hat{\imath} + \hat{\jmath})$

PUV046

16. The magnitude of vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ in the given figure are equal. The direction of $\overrightarrow{OA} + \overrightarrow{OB} - \overrightarrow{OC}$ with x-axis will be :- **[JEE (Main) 2021]**

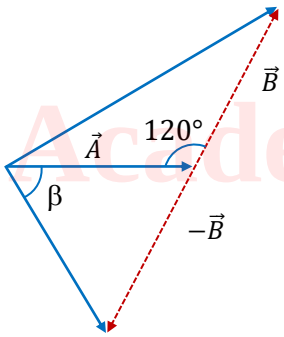


(1) $\tan^{-1}\frac{(1-\sqrt{3}-\sqrt{2})}{(1+\sqrt{3}+\sqrt{2})}$ (2) $\tan^{-1}\frac{(\sqrt{3}-1+\sqrt{2})}{(1+\sqrt{3}-\sqrt{2})}$

(3) $\tan^{-1}\frac{(\sqrt{3}-1+\sqrt{2})}{(1-\sqrt{3}+\sqrt{2})}$ (4) $\tan^{-1}\frac{(1+\sqrt{3}-\sqrt{2})}{(1-\sqrt{3}-\sqrt{2})}$

PUV048

17. The angle between vector $(\vec{A})$ and $(\vec{A} - \vec{B})$ is : **[JEE (Main) 2021]**



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(1) $\tan^{-1}\left(\frac{-\frac{B}{2}}{A - B\frac{\sqrt{3}}{2}}\right)$ (2) $\tan^{-1}\left(\frac{A}{0.7B}\right)$

(3) $\tan^{-1}\left(\frac{\sqrt{3}B}{2A - B}\right)$ (4) $\tan^{-1}\left(\frac{B\cos\theta}{A - B\sin\theta}\right)$

PUV049

18. If force (F), velocity (V) and time (T) are considered as fundamental physical quantity, then dimensional formula of density will be: **[JEE (Main) 2023]**

(1) $FV^{-2}T^{2}$ (2) $FV^{-4}T^{2}$ (3) $FV^{4}T^{-6}$ (3) $F^{2}V^{-2}T^{6}$

PUV163

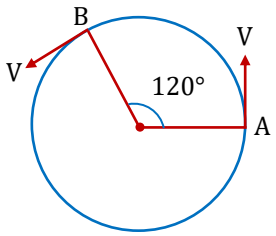
19. The speed of a wave produced in water is given by $\upsilon = \lambda^{a} g^{b} \rho^{c}$. Where λ, g and ρ are wavelength of wave, acceleration due to gravity and density of water respectively. The values of a, b and c respectively, are : **[JEE (Main) 2023]**

(1) $\frac{1}{2}, \frac{1}{2}, 0$ (2) 1, 1, 0 (3) 1, –1, 0 (4) $\frac{1}{2}, 0, \frac{1}{2}$

PUV164

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20. As shown in the figure, a particle is moving with constant speed $\pi\ m/s$. Considering its motion from A to B, the magnitude of the average velocity is: **[JEE (Main) 2023]**



(1) $\pi\ m/s$ (2) $\sqrt{3}\ m/s$ (3) $2\sqrt{3}\ m/s$ (4) $1.5\sqrt{3}\ m/s$

PUV165

21. A particle is moving with constant speed in a circular path. When the particle turns by an angle 90°, the ratio of instantaneous velocity to its average velocity is $\pi : x\sqrt{2}$. The value of x will be **[JEE (Main) 2023]**

(1) 2 (2) 5 (3) 1 (4) 7

PUV166

22. Match List-I with List-II.

	List-I		List-II
A.	Coefficient of viscosity	I.	$[M\,L^2T^{-2}]$
B.	Surface Tension	II.	$[M\,L^2T^{-1}]$
C.	Angular momentum	III.	$[M\,L^{-1}T^{-1}]$
D.	Rotational kinetic energy	IV.	$[M\,L^0T^{-2}]$

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[JEE (Main) 2024]

(1) A–II, B–I, C–IV, D–III (2) A–I, B–II, C–III, D–IV
(3) A–III, B–IV, C–II, D–I (4) A–IV, B–III, C–II, D–I

PUV168

23. Two forces $\vec{F}_1$ and $\vec{F}_2$ are acting on a body. One force has magnitude thrice that of the other force and the resultant of the two forces is equal to the force of larger magnitude. The angle between $\vec{F}_1$ and $\vec{F}_2$ is $\cos^{-1}\left(\frac{1}{n}\right)$. The value of $|n|$ is _____. **[JEE (Main) 2024]**

PUV170

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EXERCISE - JEE (Advanced) PYQ

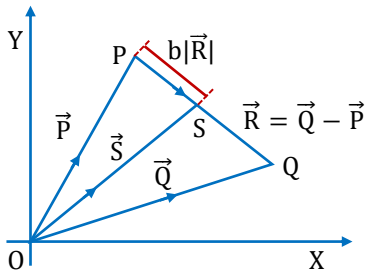
1. In terms of potential difference V, electric current I, permittivity ε_0, permeability μ_0 and speed of light c, the dimensionally correct equation(s) is(are)– **[JEE (Advanced) 2015]**

(A) $\mu_0 I^2 = \varepsilon_0 V^2$ (B) $\varepsilon_0 I = \mu_0 V$ (C) $I = \varepsilon_0 cV$ (D) $\mu_0 cI = \varepsilon_0 V$

PUV071

2. Three vectors $\vec{P}, \vec{Q}$ and $\vec{R}$ are shown in the figure. Let S be any point on the vector $\vec{R}$. The distance between the points P and S is $b|\vec{R}|$. The general relation among vectors $\vec{P}, \vec{Q}$ and $\vec{S}$ is:

[JEE (Advanced) 2017]



(A) $\vec{S} = (1-b)\vec{P} + b^2\vec{Q}$ (B) $\vec{S} = (b-1)\vec{P} + b\vec{Q}$

(C) $\vec{S} = (1-b)\vec{P} + b\vec{Q}$ (D) $\vec{S} = (1-b^2)\vec{P} + b\vec{Q}$

PUV072

3. Two vectors $\vec{A}$ and $\vec{B}$ are defined as $\vec{A} = a\hat{i}$ and $\vec{B} = a(\cos\omega t\hat{i} + \sin\omega t\hat{j})$, where a is a constant and $\omega = \pi/6$ rad s^{-1}. If $|\vec{A} + \vec{B}| = \sqrt{3}|\vec{A} - \vec{B}|$ at time t = τ for the first time, the value of τ, in seconds, is ____. **[JEE (Advanced) 2018]**

PUV073

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PARAGRAPH FOR QUESTION NO. 4 AND 5

In electromagnetic theory, the electric and magnetic phenomena are related to each other. Therefore, the dimensions of electric and magnetic quantities must also be related to each other. In the questions below, $[E]$ and $[B]$ stand for dimensions of electric and magnetic fields respectively, while $[\in_0]$ and $[\mu_0]$ stand for dimensions of the permittivity and permeability of free space respectively. $[L]$ and $[T]$ are dimensions of length and time respectively. All the quantities are given in SI units. **[JEE (Advanced) 2018]**

4. The relation between $[E]$ and $[B]$ is :-

(A) $[E] = [B][L][T]$ (B) $[E] = [B][L]^{-1}[T]$ (C) $[E] = [B][L][T]^{-1}$ (D) $[E] = [B][L]^{-1}[T]^{-1}$

PUV074

5. The relation between $[\in_0]$ and $[\mu_0]$ is :-

(A) $[\mu_0] = [\in_0][L]^2[T]^{-2}$ (B) $[\mu_0] = [\in_0][L]^{-2}[T]^2$

(C) $[\mu_0] = [\in_0]^{-1}[L]^2[T]^{-2}$ (D) $[\mu_0] = [\in_0]^{-1}[L]^{-2}[T]^2$

PUV075

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6. Let us consider a system of units in which mass and angular momentum are dimensionless. If length has dimension of L, which of the following statement (s) is/are correct ? **[JEE (Advanced) 2019]**

(A) The dimension of force is L^{-3} (B) The dimension of energy is L^{-2}

(C) The dimension of power is L^{-5} (D) The dimension of linear momentum is L^{-1}

PUV076

7. Sometimes it is convenient to construct a system of units so that all quantities can be expressed in terms of only one physical quantity. In one such system, dimensions of different quantities are given in terms of a quantity X as follows: [position] = $[X^{\alpha}]$; [speed] = $[X^{\beta}]$; [acceleration] = $[X^{p}]$; [linear momentum] = $[X^{q}]$; [force] = $[X^{r}]$. Then **[JEE (Advanced) 2020]**

(A) $\alpha + p = 2\beta$ (B) $p + q - r = \beta$ (C) $p - q + r = \alpha$ (D) $p + q + r = \beta$

PUV077

8. A physical quantity $\vec{S}$ is defined as $\vec{S} = (\vec{E} \times \vec{B}) / \mu_0$, where $\vec{E}$ is electric field, $\vec{B}$ is magnetic field and μ_0 is the permeability of free space. The dimensions of $\vec{S}$ are the same as the dimensions of which of the following quantity/quantities ? **[JEE (Advanced) 2021]**

(A) $\dfrac{\text{Energy}}{\text{Charge} \times \text{current}}$ (B) $\dfrac{\text{Force}}{\text{Length} \times \text{Time}}$

(C) $\dfrac{\text{Energy}}{\text{Volume}}$ (D) $\dfrac{\text{Power}}{\text{Area}}$

PUV078

9. In a particular system of units, a physical quantity can be expressed in terms of the electric charge e, electron mass m_e, Planck's constant h, and Coulomb's constant $k = \dfrac{1}{4\pi\epsilon_0}$, where ϵ_0 is the permittivity of vacuum. In terms of these physical constants, the dimension of the magnetic field is $[B] = [e]^{\alpha} [m_e]^{\beta} [h]^{\gamma} [k]^{\delta}$. The value of $\alpha + \beta + \gamma + \delta$ is _____. **[JEE (Advanced) 2022]**

PUV079

10. Young's modulus of elasticity Y is expressed in terms of three derived quantities, namely, the gravitational constant G, Planck's constant h and the speed of light c, as $Y = c^{\alpha} h^{\beta} G^{\gamma}$. Which of the following is the correct option ? **[JEE (Advanced) 2023]**

(A) $\alpha = 7, \beta = -1, \gamma = -2$ (B) $\alpha = -7, \beta = -1, \gamma = -2$

(C) $\alpha = 7, \beta = -1, \gamma = 2$ (D) $\alpha = -7, \beta = 1, \gamma = -2$

PUV167

11. A dimensionless quantity is constructed in terms of electronic charge e, permittivity of free space ε_0, Planck's constant h, and speed of light c. If the dimensionless quantity is written as $e^{\alpha}\varepsilon_0^{\beta}h^{\gamma}c^{\delta}$ and n is a non-zero integer, then $(\alpha, \beta, \gamma, \delta)$ is given by **[JEE (Advanced) 2024]**

(A) $(2n, -n, -n, -n)$ (B) $(n, -n, -2n, -n)$ (C) $(n, -n, -n, -2n)$ (D) $(2n, -n, -2n, -2n)$

PUV171

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	D	C	B	A	A	A	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	D	B	C	A	B	A	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	C	B	A	C	C	B	D	C	D
Que.	31	32	33	34	35					
Ans.	B	C	C	A	C					

EXERCISE (O-2)

Que.	1	2	3	4	5	6
Ans.	BC	ACD	CD	AD	A	B

EXERCISE (S)

1. $ML^5T^{-2}K^{1/2}$ **2.** 2 **3.** 2 **4.** $\sqrt{75}$

5. 1 **6.** 25 **7.** 3 **8.** $\frac{7}{16}$

9. $\hat{\imath}+\hat{\jmath}, 3\hat{k}$ **10.** 2

11. (a) $11\hat{\imath}+5\hat{\jmath}-7\hat{k}$, (b) $\cos^{-1}\left(\frac{-7}{\sqrt{195}}\right)$, (c) $\cos^{-1}\left(\frac{-20}{\sqrt{1309}}\right)$

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	3	3	2	4	4	2	4	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	180	4	4	1	3	2	1	4
Que.	21	22	23							
Ans.	1	3	6							

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AC	C	2 [1.99, 2.01]	C	D	ABD	AB	BD	4	A
Que.	11									
Ans.	A									

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